

IS 471 Semester Project:

Predicting Bank Term Deposit Purchases

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Introduction

The purpose of this project is to predict which bank customers are most likely to purchase a term deposit during a marketing campaign. Term deposits are profitable for banks because customers commit funds for a fixed period, allowing the bank to lend these funds at a higher interest rate. However, marketing these products can be costly, so the bank wants to focus its marketing resources on customers most likely to purchase.

To address this problem, I applied two analytical techniques from this course:

1. **Logistic Regression** – a regression-based classification method that predicts the probability that a customer will purchase.
2. **Decision Tree Classification** – a tree-based classifier that predicts purchase as a categorical yes/no outcome.

This project uses a data analytics approach that incorporates both statistical modeling and machine-learning techniques. Logistic Regression is a statistical model because it is based on probability theory and estimates relationships between predictors and purchase likelihood. The Decision Tree model, however, falls under data analytics and machine learning because it identifies patterns by recursively splitting the data into decision rules rather than relying on statistical assumptions. Using both methods together provides a strong combination of statistical reasoning and machine-learning insights for predicting customer behavior.

Dataset Description

The dataset contains **4521 customers** with **16 variables**, including:

- **Demographics:** age, job, marital status, education
- **Financial info:** balance, loans
- **Campaign behavior:** contact month, number of contacts, previous outcomes
- **Target variable:** Whether the customer purchased a term deposit (“yes” or “no”)

There is a strong class imbalance: 4000 customers did not purchase, while 521 purchased. No missing values were found.

Data Preprocessing

The following steps were completed before analysis:

- One-hot encoding converted categorical variables into dummy variables.
- drop_first=True was used to avoid multicollinearity.
- The target variable was encoded (no = 0, yes = 1).
- An 80/20 train-test split allowed unbiased model evaluation.

Logistic Regression Model

Top 5 Positive Predictors (These increase the chance of purchase.)

Feature	Coefficient	Interpretation
poutcome_success	2.244	successful previous campaign increase likelihood
month_oct	1.128	October is a strong purchase month
month_mar	0.881	March increases likelihood.
job_retired	0.601	Retired customers are more responsive
poutcome_other	0.583	Other positive outcomes increase probability.

Top 5 Negative Predictors (These decrease the chance of purchase.)

Feature	Coefficient	Interpretation
contact_unknown	-1.021	Unknown contact methods significantly harms success.
month_jan	-0.832	January is a weak month
job_unemployed	-0.666	Lower buying likelihood
month_may	-0.638	Historically poor month
month_nov	-0.635	Ineffective month.

Model Performance -Training Accuracy: 0.892

Decision Tree Classification Model

Confusion Matrix (Test Set):

	Predict No	Predict Yes
Actual No	3191	0
Actual Yes	423	0

The tree predicted every customer as a non-purchaser. This reflects the strong class imbalance and shows that the model defaults to the majority class instead of identifying true purchasers

This behavior demonstrates significant overfitting. Although the model performs perfectly on the training data, its test performance drops sharply and it predicts “no purchase” for every customer. This happened because the dataset is highly imbalanced, with far more non-purchasers than purchasers. As a result, the tree defaults to predicting the majority class, which inflates accuracy but provides almost no real predictive value. This confirms that while decision trees are easy to interpret, they can perform poorly without additional tuning or handling of class imbalance.

Model Accuracy

- Training Accuracy: 1.00
- Test Accuracy: 0.808

Top 5 Influential Features (from Decision Tree):

Feature	Importance
balance	0.194016
age	0.161213
day	0.133275
poutcome_success	0.086328
campaign	0.057844

Model Comparison and Interpretation

1. Performance

- Logistic Regression: **0.892 accuracy**, more stable.
- Decision Tree: **0.808 accuracy**, but overfits (1.00 training).

Logistic regression generalizes better and provides more reliable patterns.

2. Agreement on Predictors

Both models highlight:

- previous campaign success
- timing (month)
- financial factors (balance)
- demographics such as age and job

This consistency increases confidence in the insights.

3. Model Strengths

- **Logistic Regression:** Best for probability-based targeting.
- **Decision Tree:** Useful for simple decision rules, but struggles with imbalance data.

Insights and Business Recommendations

The purpose of these models is not only to predict which customers may purchase but also to understand why certain customers are more likely to buy. The following insights interpret the statistical and model results into meaningful business recommendations.

Based on the results:

- 1. Prioritize customers with previous campaign success.** - They show the highest likelihood of purchase.
- 2. Run campaigns in strong months (October, March)** - Avoid January, May, and November.
- 3. Target customers with strong financial profiles** - Balance is one of the strongest predictors.
- 4. Older customers and retirees are more responsive**
- 5. Improve contact method accuracy** - Unknown contact methods significantly reduce success.

Conclusion

Both logistic regression and the decision tree identified meaningful predictors of term deposit purchase behavior. Logistic regression proved to be more stable and accurate, especially with the imbalanced dataset, while the decision tree provided helpful interpretability even though it struggled on the test set. Overall, the results suggest that the bank should focus its marketing efforts on customers with successful prior engagement, stronger financial profiles, and strategic timing. Applying these insights can help reduce marketing costs and increase conversion rates.

Appendix A: AI Assistance Disclosure

AI Tool Used:

ChatGPT by OpenAI (GPT-5.1, ChatGPT Plus)

Purpose of Use:

AI assistance was used only for writing support, clarity improvements, grammar editing, and organization of my final report.

All coding, statistical work, model building, data preprocessing, evaluation, and interpretation of results were completed independently by me.

Prompts / Instructions Provided to the AI

Below is a summary of the types of prompts I used:

- Requests to rephrase, organize, or clarify my written explanations
- Requests to clean up grammar while keeping my tone
- Requests to check that my description of logistic regression and decision trees was correct
- Requests to help with formatting my conclusion and introduction
- Requests to verify that my interpretation of confusion matrices and model performance was stated clearly
- Requests to help determine where to insert certain paragraphs

AI-Generated Output Included in Report

AI-generated text contributed to the following areas:

- Minor rewriting and polishing of the **Introduction, Model Comparison, Insights, and Conclusion**
- Suggestions for wording and structure of explanation paragraphs
- Grammar correction throughout the report

All final text was reviewed, edited, and approved by me before inclusion.